



Islamic University of Technology
Department of Computer Science and Engineering

Lab 4: Shortest Path

CSE 4404: Algorithms Lab
Summer 2024-25

Contents

Task A. Shortest Path	2
Task Statement	2
Input Format	2
Output Format	2
Examples	2
Task B. Maximum Probability Path	3
Task Statement	3
Input Format	3
Output Format	3
Examples	3
Task C. Jzzhu and Cities	4
Task Statement	4
Input Format	4
Output Format	4
Examples	4

Task A. Shortest Path

Time Limit: 1.0 second | Memory Limit: 512 MB

You are given a weighted undirected graph with n vertices enumerated from 1 to n . Your task is to find the shortest path between vertex 1 and vertex n .

If there are multiple shortest paths, any of them will be accepted. Note that the graph may contain loops and multiple edges between the same pair of vertices.

Input Format

The first input line contains two integers n and m : the number of vertices and the number of edges.

After this, there are m lines describing the edges. Each line contains three integers a_i , b_i , and w_i : an undirected edge exists between vertex a_i and vertex b_i with weight w_i .

Output Format

If no path exists, print -1 .

Otherwise, print the sequence of vertices representing the shortest path from vertex 1 to vertex n .

Constraints

$$2 \leq n \leq 10^5$$

$$0 \leq m \leq 10^5$$

$$1 \leq a_i, b_i \leq n$$

$$1 \leq w_i \leq 10^6$$

Examples

Sample Input	Sample Output
5 6 1 2 2 2 5 5 2 3 4 1 4 1 4 3 3 3 5 1	1 4 3 5
3 1 1 2 5	-1

Task B. Maximum Probability Path

Time Limit: 1.0 second | Memory Limit: 512 MB

You are given an undirected graph with n vertices (enumerated from 0 to $n - 1$) and m edges. Each edge has an associated probability of success w ($0 \leq w \leq 1$).

Your task is to find a path from a specified start node to an end node that results in the highest possible product of success probabilities along that path.

If no path exists between the two nodes, the probability is considered to be 0.

Input Format

The first input line contains two integers n and m : the number of vertices and the number of edges.

The next m lines describe the edges. Each line contains two integers u, v and a floating-point number w : an undirected edge exists between vertex u and vertex v with a success probability w .

The final line contains two integers $start$ and end : the source and destination nodes.

Output Format

Print a single floating-point number: the maximum probability of reaching the end node from the $start$ node.

The answer is considered correct if its absolute or relative error does not exceed 10^{-5} .

Constraints

$$\begin{aligned} 2 &\leq n \leq 10^4 \\ 0 &\leq m \leq 2 \cdot 10^4 \\ 0 &\leq u, v < n \\ 0 &\leq w \leq 1 \\ 0 &\leq start, end < n \end{aligned}$$

Examples

Sample Input	Sample Output
3 3 0 1 0.5 1 2 0.5 0 2 0.2 0 2	0.25000

Task C. Jzzhu and Cities

Time Limit: 2 seconds | Memory Limit: 256 MB

Jzzhu is the president of a country, and he wants to build a country with developed transportation.

There are n cities in his country, and m bidirectional roads. Each road connects two different cities and has a positive integer length. There can be multiple roads between two cities. The capital of the country is city 1.

Also, there are k train routes. Each train route connects the capital with some other city s_i and has length y_i . Each train route is one-way, and there can be multiple train routes to the same city, possibly with different lengths.

Jzzhu thinks some of the train routes are unnecessary. A train route is unnecessary if there exists a path from the capital to the destination city using only roads and/or other train routes with strictly smaller total length.

Your task is to determine how many train routes are unnecessary.

Input Format

The first line contains three integers n , m , and k ($2 \leq n \leq 10^5$, $1 \leq m \leq 10^5$, $1 \leq k \leq 10^5$).

The next m lines each contain three integers u , v , and x ($1 \leq u, v \leq n$, $u \neq v$, $1 \leq x \leq 10^9$) — a road between cities u and v with length x .

Then k lines follow. Each of them contains two integers s and y ($2 \leq s \leq n$, $1 \leq y \leq 10^9$) — meaning a train route from the capital (city 1) to city s with length y .

Output Format

Print a single integer — the number of unnecessary train routes.

Examples

Sample Input	Sample Output
5 5 3 1 2 1 2 3 2 1 3 3 3 4 4 1 5 5 3 5 4 5 5 5	2
2 2 3 1 2 2 2 1 3 2 1 2 2 2 3	2

Marks Distribution

Task	Marks
Task A	40%
Task B	30%
Task C	30%

Practice Problems

Problem	Links
Shortest Path	Library Checker: https://judge.yosupo.jp/problem/shortest_path Vjudge: https://vjudge.net/problem/Yosupo-shortest_path
Dijkstra?	Codeforces: https://codeforces.com/problemset/problem/20/C Vjudge: https://vjudge.net/problem/CodeForces-20C
Jzzhu and Cities	Codeforces: https://codeforces.com/problemset/problem/449/B Vjudge: https://vjudge.net/problem/CodeForces-449B
Graph and Graph	Codeforces: https://codeforces.com/contest/2059/problem/D Vjudge: https://vjudge.net/problem/CodeForces-2059D